Maths assignment - 5

1. If $\bar{F} = 3xy\bar{i} - y^2\bar{j}$, evaluate $\int_C \bar{F}.d\bar{r}$ where C is the curve $y = 2x^2$ in xy-plane from (0,0) to (1,2).

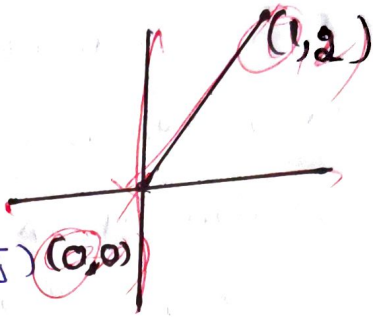
Given $\bar{F} = 3xy\bar{i} - y^2\bar{j}$,

the curve $y = 2x^2$ in xy-plane then $z = 0$.

$$\bar{r} = x\bar{i} + y\bar{j} + z\bar{k}$$

$$= x\bar{i} + y\bar{j} + (0)\bar{k}$$

$$d\bar{r} = dx\bar{i} + dy\bar{j}$$



$$\bar{F}.d\bar{r} = (3xy\bar{i} - y^2\bar{j}).(dx\bar{i} + dy\bar{j})$$

$$= 3xy\,dx - y^2dy$$

given $y = 2x^2 \Rightarrow dy = 4x\,dx$

$$\bar{F}.d\bar{r} = 3x(2x^2)dx - (2x^2)^2(4x\,dx)$$

$$= (6x^3 - 16x^5)dx$$

Also $x = (0,0)$ to $(1,2)$

$x = 0$ to 1

Now, $\int_{x=0}^{1} (6x^3 - 16x^5)dx$

$$= 6\cdot\frac{x^4}{4} - 16\frac{x^6}{6}\Big]_0^1$$

$$= \left[\frac{6}{4}(1) - \frac{16}{6}(1) - 0\right]$$

$$= \frac{36 - 64}{24}$$

$$= \frac{-28}{24}$$

$$= \frac{-7}{6}$$

2. find the workdone in moving a particle in the force field $\bar{F} = 3x^2\bar{i} + (2xz-y)\bar{j} + z\bar{k}$ along the straight line from (0,0,0) to (2,1,3)

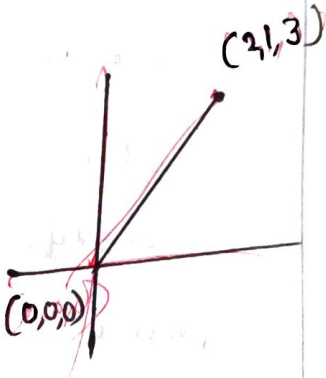
Soln

Given, $\bar{F} = 3x^2\bar{i} + (2xz-y)\bar{j} + z\bar{k}$

$\det \bar{r} = x\bar{i} + y\bar{j} + z\bar{k}$, $\bar{dr} = dx\bar{i} + dy\bar{j} + dz\bar{k}$

$\bar{F}\cdot\bar{dr} = 3x^2dx + (2xz-y)dy + zdz$

Let O = (0,0,0) and A (2,1,3)



equation of OA is,

$$\frac{x-0}{2-0} = \frac{y-0}{1-0} = \frac{z-0}{3-0} = t$$

$\Rightarrow x = 2t, \quad y = t, \quad z = 3t$

$dx = 2dt, \quad dy = dt, \quad dz = 3dt$

limits $t = 0$ to 1

$\bar{F}\cdot\bar{dr} = 3(2t)^2(2dt) + (2(2t)(3t)-t)dt + 3t(3dt)$

$= (24t^2 + 12t^2 - t + 9t)\,dt$

$= (36t^2 + 8t)\,dt$

Total workdone $= \int_{OA} \bar{F}\cdot d\bar{r}$

$= \int_{t=0}^{1} (36t^2 + 8t)\,dt$

$= 36\frac{t^3}{3} + 8\frac{t^2}{2}\Big]_0^1$

$= (12 + 4) - 0$

$= 16$

find the workdone, by force $\bar{F} = z\bar{i} + x\bar{j} + y\bar{k}$ when it moves along the curve $\bar{r} = \cos t\,\bar{i} + \sin t\,\bar{j} - t\bar{k}$ from $t = 0$ to 2π.

Given, curve $\bar{r} = \cos t\,\bar{i} + \sin t\,\bar{j} - t\bar{k}$

Since, $\bar{r} = x\bar{i} + y\bar{j} + z\bar{k}$

$x = \cos t, \quad y = \sin t, \quad z = -t$

$\bar{r} = \cos t\,\bar{i} + \sin t\,\bar{j} - t\bar{k}$

$d\bar{r} = (-\sin t\,\bar{i} + \cos t\,\bar{j} - \bar{k})\,dt$

$\bar{F} \cdot d\bar{r} = (z\bar{i} + x\bar{j} + y\bar{k}) \cdot (-\sin t\,\bar{i} + \cos t\,\bar{j} - \bar{k})\,dt$

$= (-z\sin t + x\cos t - y)\,dt$

$= (-(-t)\sin t + \cos t \cdot \cos t - \sin t)\,dt$

$= (t\sin t + \cos^2 t - \sin t)\,dt$

Work done from $t = 0$ to 2π

$$w = \int_{t=0}^{2\pi} F \cdot d\bar{r} = \int_{t=0}^{2\pi} (t\sin t + \cos^2 t - \sin t)\,dt$$

$$= \int_{t=0}^{2\pi} (t\sin t + \cos^2 t - \sin t)\,dt$$

$$= \left[t(-\cos t) - 1(\sin t)\right]_0^{2\pi} + \int_{t=0}^{2\pi} \cos^2 t\,dt - \left[(-\cos t)\right]_0^{2\pi}$$

$$= -t\cos t + \sin t\Big]_0^{2\pi} + \int_{t=0}^{2\pi} \frac{1}{2}(1 + \cos 2t)\,dt + (\cos t)_{t=0}^{2\pi}$$

$$= -2\pi(-1)^2 + 0 - (0+0) + \frac{1}{2}\left[2\pi + \frac{\sin 2(2\pi)}{2} - \left(0 + \frac{\sin 0}{2}\right)\right] + (-1)^2 - 1$$

$$= -2\pi + \pi$$

$$= -\pi$$

(4) find $\int_S \bar{F}\cdot\bar{n}\,ds$ where $\bar{F} = 12x^2y\,\bar{i} - 3yz\,\bar{j} + 2z\,\bar{k}$ and s is the portion of the plane $x+y+z=1$ included in the first octant.

Soln

Given $\bar{F} = 12x^2y\,\bar{i} - 3yz\,\bar{j} + 2z\,\bar{k}$

$\phi : x+y+z-1$

$\nabla\phi = \left(\bar{i}\frac{\partial}{\partial x} + \bar{j}\frac{\partial}{\partial y} + \bar{k}\frac{\partial}{\partial z}\right)(x+y+z-1)$

$= \bar{i}+\bar{j}+\bar{k}$

$\bar{n} = \frac{\nabla\phi}{|\nabla\phi|} = \frac{\bar{i}+\bar{j}+\bar{k}}{\sqrt{3}} = \frac{1}{\sqrt{3}}(\bar{i}+\bar{j}+\bar{k})$

Now, $\bar{F}\cdot\bar{n} = (12x^2y\,\bar{i} - 3yz\,\bar{j} + 2z\,\bar{k})\frac{1}{\sqrt{3}}(\bar{i}+\bar{j}+\bar{k})$

$= \frac{1}{\sqrt{3}}(12x^2y - 3yz + 2z)$ $\{\because z = (1-x-y)\}$

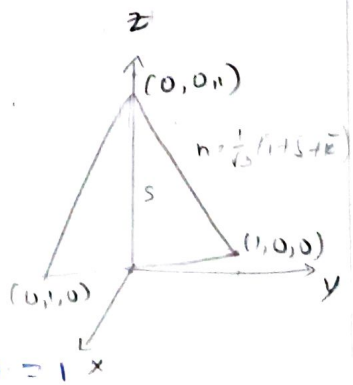
$= \frac{1}{\sqrt{3}}(12x^2y - 3y + 3xy + 3y^2 + 2 - 2x - 2y)$

$= \frac{1}{\sqrt{3}}(12x^2y + 3y^2 + 3xy - 2x - 5y + 2)$

on xy plane $ds = \frac{dx\,dy}{|\bar{n}\cdot\bar{k}|} = \frac{dx\,dy}{\left|\left(\frac{\bar{i}+\bar{j}+\bar{k}}{\sqrt{3}}\right)\cdot\bar{k}\right|}$

$= \frac{dx\,dy}{\frac{1}{\sqrt{3}}}$

$= \sqrt{3}\,dx\,dy$



since given plane $x+y+z=1$

on xy plane $z=0$, $\therefore x+y+0=1$

$y = 1-x$

in the first octant $y=0$ to $1-x$

and $x+0+0=1 \Rightarrow x=1$

∴ But octant $x=0$ to 1

Now, $\int_S \bar{F}\cdot\bar{n}\,ds = \iint_R \frac{1}{\sqrt{3}}(12x^2y+3y^2+3xy-2x-5y+2)\sqrt{3}\,dx\,dy$

$$= \int_{x=0}^{1}\int_{y=0}^{1-x}(12x^2y+3y^2+3xy-2x-3y+2)\,dy\,dx$$

$$= \int_{x=0}^{1}\left[12x^2\frac{y^2}{2}+3\frac{y^3}{3}+3x\frac{y^2}{2}-2xy-3\frac{y^2}{2}+2y\right]_0^{1-x}dx$$

$$= \int_0^1\left[6x^2(1-x^2)+(1-x)^3+\frac{3}{2}x(1-x^2)-2x(1-x)-\frac{5}{2}(1-x)^2+2(1-x)-0\right]dx$$

$$= \int_0^1(1-x)\left[6x^2-6x^3+1+x^2-2x+\frac{3}{2}x^2-2x-\frac{5}{2}+\frac{5}{2}x+2\right]dx$$

$$= \int_0^1(1-x)\left(-6x^3+\frac{11}{2}x^2+\frac{1}{2}\right)dx$$

$$= \int_0^1\left(-6x^3+\frac{11}{2}x^2+\frac{1}{2}+6x^4-\frac{11}{2}x^3-\frac{1}{2}x\right)dx$$

$$= -6\frac{x^4}{4}+\frac{11x^3}{6}+\frac{x}{2}+\frac{6x^5}{5}-\frac{11x^4}{8}-\frac{x^2}{4}\Big]_0^1$$

$$= \left[\frac{-3}{2}+\frac{11}{6}+\frac{1}{2}+\frac{6}{5}-\frac{11}{8}-\frac{1}{4}\right]$$

$$= \left[-1-\frac{13}{8}+\frac{11}{6}+\frac{6}{5}\right]$$

$$= -\frac{22}{8}+\frac{41}{30}$$

$$= \frac{49}{120}$$

(5) Evaluate $\int \bar{F}\cdot\bar{n}\,ds$ where $\bar{F} = z\bar{i} + x\bar{j} - 3y^2z\bar{k}$ and S is the surface $x^2+y^2=16$ included the first octant between $z=0$ and $z=5$

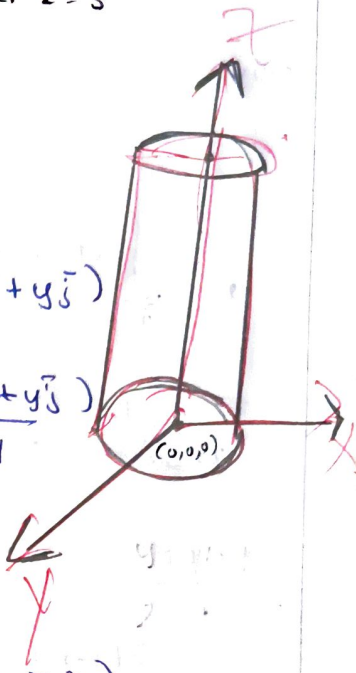
Soln

Given $\bar{F} = z\bar{i} + x\bar{j} - 3y^2z\bar{k}$

Let $\phi = x^2+y^2-16$ then

normal $= \nabla\phi = 2x\bar{i} + 2y\bar{j} = 2(x\bar{i}+y\bar{j})$

$$\bar{n} = \frac{2(x\bar{i}+y\bar{j})}{2\sqrt{x^2+y^2}} = \frac{x\bar{i}+y\bar{j}}{\sqrt{16}} = \frac{(x\bar{i}+y\bar{j})}{4}$$



$$\int_S \bar{F}\cdot\bar{n}\,ds = \iint \bar{F}\cdot\bar{n}\,\frac{dydz}{|\bar{n}\cdot i|}$$

$$\bar{F}\cdot\bar{n} = (z\bar{i}+x\bar{j}-3y^2z\bar{k})\cdot\left(\frac{1}{4}(x\bar{i}+y\bar{j})\right)$$

$$= \frac{1}{4}(xz+xy)$$

$$\bar{n}\cdot\bar{i} = \frac{1}{4}(x\bar{i}+y\bar{j})\cdot\bar{i}$$

$$= \frac{x}{4}$$

for the surface $x^2+y^2=16$ in the yz-plane

$x=0 \Rightarrow y=4$

$y=0$ to 4 and $z=0$ to 5

$$\int \bar{F}\cdot\bar{n}\,ds = \int_S \bar{F}\cdot\bar{n}\,\frac{dydz}{|\bar{n}\cdot i|}$$

$$= \iint \left(\frac{xz+xy}{4}\right)\left(\frac{4}{x}\right)dydz$$

$$= \int_{y=0}^{4}\int_{z=0}^{5} (y+z)\,dy\,dz$$

$$= \int_{z=0}^{5} \left[\frac{y^2}{2} + 2y\right]_0^4 dz$$

$$= \int_0^5 \left(\frac{16}{2} + 4z\right) dx$$

$$= \left[\frac{16z}{2} + 2z^2\right]_0^5$$

$$= 40 + 50 = 90$$

(6) Evaluate $\int_V \bar{F} \, dv$, given that $\bar{F} = 2xz\,\bar{i} - x\,\bar{j} + y^2\,\bar{k}$ and V is the region bounded by the surfaces $x = 0$, $x = 2$; $y = 0$, $y = 6$ and $z = x^2$, $z = 4$

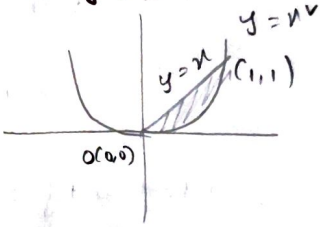
(7) Verify green's theorem for $\int_C [(xy + y^2)dx + x^2 dy]$, where C is bounded by $y = x$ and $y = x^2$.

given curve $y = x$ & $y = x^2$

$x^2 = x \Rightarrow x(x-1) = 0$

$x = 0 \Rightarrow y = x = 0$

$x = 1 \Rightarrow y = x = 1$



∴ points O(0,0) & A(1,1)

Green's theorem states that

$$\int_C M dx + N dy = \iint_S \left[\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right] dx\,dy$$

To find $\int M dx + N dy$; $\Rightarrow \int_C M dx + N dy = \int_{OA} M dx + N dy + \int_A M dx + N dy$

$M = x + y^2$, $N = x^2$ Along OA : $y = x^2$, $dy = 2x\,dx$

$$\int_{OA} M dx + N dy = \int (xy + y^2) dx + x^2 2x\,dx \qquad x = 0 \text{ to } 1$$

$$= \int_{x=0}^{1} \left[x x^2 + (x^2)^2\right] dx + x^2 2x\,dx$$

$$= \int_0^1 (x^3 + x^4 + 2x^3)\, dx$$

$$= \int_0^1 (x^4 + 3x^3)\,dx$$

$$= \frac{x^5}{5} + 3\frac{x^4}{4}\Big]_0^1$$

$$\int_{OA} M\,dx + N\,dy = \left[\frac{1}{5} + \frac{3}{4}\right] = \frac{19}{20}$$

along AO : $y = x$ then $dy = dx$

$x = 1$ to 0

$$\int_{AO} M\,dx + N\,dy = \int_0^1 (xy + y^2)\,dx + x^2\,dy$$

$$= \int_1^0 (x \cdot x + x^2)\,dx + x^2\,dx$$

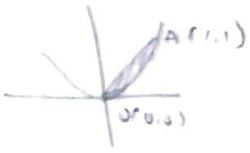
$$= \int_1^0 3x^2\,dx \quad \Rightarrow 3\left[\frac{x^3}{3}\right]_1^0$$

$$\Rightarrow 0 - 1^3 = -1$$

$$\int_C M\,dx + N\,dy = \frac{19}{20} - 1$$

$$= \frac{-1}{20} \quad \text{— ①}$$

TO find $\iint_S \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) dx\,dy$



$M = xy + y^2 \qquad N = x^2$

$\frac{\partial M}{\partial y} = x + 2y \qquad \frac{\partial N}{\partial x} = 2x$

$$\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} = 2x - (x + 2y) = x - 2y$$

Take the strip parallel to y-axis

$y = x^2$ to x ; $x = 0$ to 1

Now $\iint_S \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) dx\,dy$

$= \int_{x=0}^{1} \int_{y=x^2}^{2} (x-2y)\, dy\, dx$

$= \int_{x=0}^{1} \left[xy - 2\frac{y^2}{2}\right]_{x^2}^{x} dx$

$= \int_{0}^{1} \left((x^2-x^2) - (x^3-x^4)\right) dx$

$= \int_{0}^{1} (x^4-x^3)\, dx = \frac{x^5}{5} - \frac{x^4}{4}\Big]_0^1$

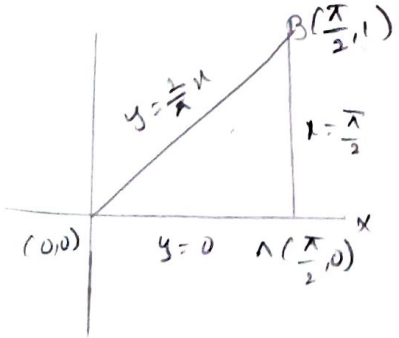
$= \frac{1}{5} - \frac{1}{4} = \frac{-1}{20}$

By ① & ②

$\int_C M dx + N dy = \iint_S \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) dx\,dy$

Green's theorem verified.

(8) verify green's theorem, evaluate $\int_C (y-\sin x)dx + \cos x\, dy$ where C is the curve enclosed by the lines $y=0$, $x=\frac{\pi}{2}$ and $\pi y = 2x$.



given curve

$y=0$, $x=\frac{\pi}{2}$, $y=\frac{2}{\pi}x$

$y=\frac{2}{\pi}x$, $x=\frac{\pi}{2}$

$y = \frac{2}{\pi}\cdot\frac{\pi}{2} = 1 \Rightarrow y = 1$

given $M = y - \sin x$, $N = \cos x$

$\frac{\partial M}{\partial y} = 1-0$ $\qquad$ $\frac{\partial N}{\partial x} = -\sin x$

$\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} = -\sin x - 1 = -(\sin x + 1)$

By green's theorem we have

$$\int M dx + N dy = \iint_S \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) dx\, dy$$

Take the strip parallel to y-axis

$y = 0$ to $\frac{2}{\pi}x$

$x = 0$ to $\frac{\pi}{2}$

$$\int M dx + N dy = \iint_S \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) dx\, dy$$

$$\int (y - x) dx + \cos x\, dy = \int_{x=0}^{\pi/2} \int_{y=0}^{\frac{2}{\pi}x} -(1 + \sin x)\, dy\, dx$$

$$= -\int_0^{\pi/2} (1 + \sin x)\, (y)_0^{\frac{2}{\pi}x}\, dx$$

$$= -\frac{2}{\pi} \int_0^{\pi/2} (x + 1)(\sin x)\, dx$$

$$= \frac{-2}{\pi} \left[\frac{x^2}{2} + x(-\cos x) + (-\sin x)\right]_0^{\pi/2}$$

$$= \frac{-2}{\pi} \left[\frac{x^2}{2} - x\cos x + \sin x\right]_{x=0}^{\pi/2}$$

$$= \frac{-2}{\pi} \left[\frac{1}{2}\left(\frac{\pi}{2}\right)^2 - \frac{\pi}{2}\cos\left(\frac{\pi}{2}\right) + \sin\frac{\pi}{2} - (0 - 0 + \sin 0)\right]$$

$$= -\frac{2}{\pi} \left[\frac{\pi^2}{8} - \frac{\pi}{2} \cdot 0 + 1 - (0)\right]$$

$$= \frac{-2}{\pi} \left(\frac{\pi^2}{8} + 1\right)$$

$$= -\left(\frac{2}{\pi}\frac{\pi^2}{8} + \frac{2}{\pi}\right)$$

$$= -\left(\frac{\pi}{4} + \frac{2}{\pi}\right).$$

11) verify Gauss Divergence theorem for $\bar{F} = (x^2-yz)\bar{i} + (y^2-zx)\bar{j} + (z^2-xy)\bar{k}$ taken over the rectangular parallelopiped $0 \le x \le a$, $0 \le y \le b$, $0 \le z \le c$

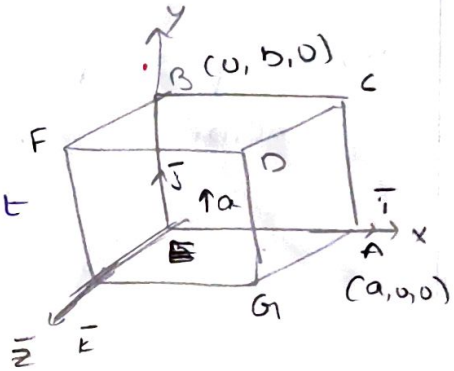
Soln

given

$$\bar{F} = (x^2-yz)\bar{i} + (y^2-zx)\bar{j} + (z^2-xy)\bar{k}$$

$x=0$, $x=a$, $y=0$, $y=b$

$z=0$, $z=c$



Gauss divergence states that

$$\int_S \bar{F}\cdot\bar{n}\, ds = \int_V \text{div}\,\bar{F}\, dv$$

TO find $\int_S \bar{F}\cdot\bar{n}\, ds$

(i) OAEB, on xy-plane $z=0$, $\bar{n} = -\bar{k}$

$$\bar{F}\cdot\bar{n} = \bar{F}\cdot(-\bar{k}) = -(z^2-xy) = -(0-xy) = xy$$

$$ds = \left[\frac{dx\,dy}{|\bar{n}\cdot\bar{k}|} = \frac{dx\,dy}{|-\bar{k}\cdot\bar{k}|}\right] = dx\,dy$$

and $x=0$ to a, $y=0$ to b

$$\int_{OAEB} \bar{F}\cdot\bar{n}\, ds = \int_{x=0}^{a}\int_{y=0}^{b} xy\, dx\, dy$$

$$= \int_0^a x\,dx \int_0^b y\,dy$$

$$= \frac{x^2}{2}\Big]_0^a \quad \frac{y^2}{2}\Big]_0^b$$

$$\int_{OAEB} \bar{F}\cdot\bar{n}\, ds = \frac{a^2b^2}{4}$$

(ii) along CGDF,

parallel to xy plane; $z = c$, $\bar{n} = \bar{k}$

$$\bar{F}.\bar{n} = \bar{F}.\bar{k} = z^2 - xy = c^2 - xy$$

$$ds = \left[\frac{dx\,dy}{|\bar{n}.\bar{k}|} = \frac{dx\,dy}{|\bar{k}.\bar{k}|}\right] = dx\,dy$$

$x = 0$ to a. & $y = 0$ to b

Now, $\int_{CGDF} \bar{F}.\bar{n}\,ds = \int_{x=0}^{a}\int_{y=0}^{b} (c^2 - xy)\,dy\,dx$

$$= \int_{x=0}^{a} c^2 y - x\frac{y^2}{2}\Big]_{y=0}^{b} dx = \int_{x=0}^{a} \left[c^2 b - x\frac{b^2}{2} - (0-0)\right] dx$$

$$= \int_{x=0}^{a} \left[bc^2 - \frac{b^2}{2}x\right] dx$$

$$= bc^2 x - \frac{b^2}{2}\frac{x^2}{2}\Big]_0^a$$

$$\int_{CGDF} \bar{F}.\bar{n}\,ds = \left[bc^2 a - \frac{b^2}{2}\cdot\frac{a^2}{2} - (0-0)\right]$$

$$= abc - \frac{a^2 b^2}{4}$$

(iii) OBFC

on yz-plane, $\bar{n} = -\bar{i}$, $ds = dy\,dz$

$$\bar{F}.\bar{n} = -(x^2 - yz) = -(0 - yz) = yz$$

$$\left[ds = \frac{dy\,dz}{|\bar{n}.\bar{i}|} = \frac{dy\,dz}{|\bar{i}.(-\bar{i})|} = dy\,dz\right]$$

$y = 0$ to b and $z = 0$ to c

$$\int_{OBFC} \bar{F}.\bar{n}\,ds = \int_{y=0}^{b}\int_{z=0}^{c} yz\,dy\,dz = \int_0^b y\,dy \int_0^c z\,dz$$

$$= \frac{y^2}{2}\Big]_0^b \left[\frac{z^2}{2}\Big]_0^b \Rightarrow \frac{b^2 c^2}{4}$$

(iv) Along AEDG: parallel to yz-plane

$x = a$, $\bar{n} = \bar{i}$, $ds = dy\,dz$

$\bar{F} \cdot \bar{n} = x^2 - yz = a^2 - yz$, $\left[ds = \frac{dy\,dz}{|\bar{n} \cdot \bar{i}|} = \frac{dy\,dz}{|\bar{i} \cdot \bar{i}|} = dy\,dz\right]$

$y = 0$ to b & $z = 0$ to c

$$\int_{AEDG} \bar{F} \cdot \bar{n}\, ds = \int_{x=0}^{b} \int_{y=0}^{c} (a^2 - yz)\, dz\, dy$$

$$= \int_{y=0}^{b} \left[a^2 z - y \frac{z^2}{2}\right]_0^c dy$$

$$= \int_0^b \left[a^2 z - y \frac{z^2}{2}\right]_0^c dy$$

$$= \int_0^b \left[a^2 c - y \frac{c^2}{2}\right] dy$$

$$= a^2 cy - \frac{y^2}{2} \frac{c^2}{2} \Big]_0^b$$

$$= a^2 cb - \frac{b^2 c^2}{4}$$

(v) Along OAGC: on xz-plane $y = 0$ & $\bar{n} = -\bar{j}$, $ds = dx\,dy$

$\bar{F} \cdot \bar{n} = \bar{F} \cdot (-\bar{j}) = -(y^2 - xz) = -(0 - xz)$

$\bar{F} \cdot \bar{n} = xz$

$\left[ds = \frac{dx\,dz}{|\bar{n} \cdot \bar{j}|} = \frac{dx\,dz}{|-\bar{j} \cdot \bar{j}|} = dx\,dy\right]$

$x = 0$ to a, $z = 0$ to c

$$\int_{OAGC} \bar{F} \cdot \bar{n}\, ds = \int_{x=0}^{a} \int_{y=0}^{c} xz\, dz\, dx = \int_0^a x\, dx \int_0^c z\, dz$$

$$= \frac{x^2}{2} \Big]_0^a \left[\frac{z^2}{2}\right]_0^c = \frac{a^2}{2} \frac{c^2}{2}$$

$$\Rightarrow \frac{a^2 c^2}{4}$$

(vi) along BEDF, parallel to xz-plane

$y = b$, $\bar{n} = \bar{j}$, $ds = dx\,dz$

$\bar{F}\cdot\bar{n} = \bar{F}\cdot\bar{j} = y^2 - xz = b^2 - xz$

$$ds = \frac{dx\,dz}{|\bar{n}\cdot\bar{j}|} = \frac{dx\,dz}{|\bar{j}\cdot\bar{j}|} = dx\,dz$$

$x = 0$ to a & $z = 0$ to c

$$\int_{BEDF} \bar{F}\cdot\bar{n}\,ds = \int_{z=0}^{c}\int_{y=0}^{a} (b^2 - xz)\,dx\,dz = \int_{z=0}^{c} \left(b^2x - \frac{x^2}{2}z\right)_0^a dz$$

$$= \int_0^c \left[b^2a - \frac{a^2}{2}z \right] dz$$

$$= b^2az - \frac{a^2}{2}\frac{z^2}{2}\Big]_0^c$$

$$\int_{BEDF} \bar{F}\cdot\bar{n}\,ds = ab^2c - \frac{a^2c^2}{4}$$

$$\int \bar{F}\cdot\bar{n}\,ds = \iint_{OAFB} \bar{F}\cdot\bar{n}\,ds + \iint_{CGDF} \bar{F}\cdot\bar{n}\,ds + \iint_{OBFC} \bar{F}\cdot\bar{n}\,ds +$$

$$\iint_{AEDG} \bar{F}\cdot\bar{n}\,ds + \iint_{OAGC} \bar{F}\cdot\bar{n}\,ds + \iint_{BEDF} \bar{F}\cdot\bar{n}\,ds$$

$$= \frac{a^2b^2}{4} + abc^2 - \frac{a^2b^2}{4} + \frac{b^2c^2}{4} + a^2b^2c - \frac{b^2c^2}{4} + \frac{a^2c^2}{4} + ab^2c - \frac{a^2c^2}{4}$$

$$= abc^2 + a^2bc + ab^2c$$

$$= abc\,(c + a + b)$$

$$\int_S \bar{F}\cdot\bar{n}\,ds = abc\,(c + a + b) \quad \text{— (1)}$$

To find : $\int_V \text{div } \bar{F}\, dV$

$$\text{div } \bar{F} = \nabla \cdot \bar{F}$$

$$= \left(\bar{i}\frac{\partial}{\partial x} + \bar{j}\frac{\partial}{\partial y} + \bar{k}\frac{\partial}{\partial z}\right)(x^2 - yz)\bar{i} + (y^2 - 2x)\bar{j} + (z^2 - xy)\bar{k}$$

$$= \frac{\partial}{\partial x}(x^2 - yz) + \frac{\partial}{\partial y}(y^2 - 2x) + \frac{\partial}{\partial z}(z - xy)$$

$$= 2x + 2y + 2z$$

$$= 2(x + y + z)$$

Now, $\int_V \text{div } \bar{F}\, dV$

$$= \int_{x=0}^{c}\int_{y=0}^{b}\int_{z=0}^{a} 2(x+y+z)\, dx\, dy\, dz$$

$$= 2\int_{x=0}^{c}\int_{y=0}^{b}\left[\frac{x^2}{2} + yx + zx\right]_0^a dy\, dz$$

$$= 2\int_0^c\int_0^b \left(\frac{a^2}{2} + ay + az\right) dy\, dz$$

$$= 2\int_0^c \left[\frac{a^2}{2}y + a\frac{y^2}{2} + azy\right]_0^b dz$$

$$= 2\int_0^c \left(\frac{a^2}{2}b + a\frac{b^2}{2} + azb\right) dz$$

$$= 2\left[\frac{a^2b}{2}z + a\frac{b^2}{2}z + a\frac{z^2}{2}b\right]_0^c$$

$$\int_V \text{div}\bar{F}\, dV = 2\left[\frac{a^2}{2}(bc) + a\frac{b^2}{2}c + a\frac{c^2}{2}b\right]$$

$$\int_V \text{div}\bar{F}\, dV = 2\,\frac{abc}{2}(a+b+c) = abc(a+b+c) \quad \text{---(2)}$$

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$$\iint_S \bar{F}\cdot\bar{n}\,ds = \int_V \operatorname{div}\bar{F}\,dv$$

Hence, Gauss-divergence theorem is verified.

12. Evaluate divergence theorem. evaluate $\iint x\,dy\,dz + y\,dz\,dx + z\,dx\,dy$ where S is $x^2+y^2+z^2=a^2$

Soln

given $\iint x\,dy\,dz + y\,dz\,dx + z\,dx\,dy$

$$\iint_S \bar{F}_1\,dx\,dz + f_2\,dx\,dz + f_3\,dx\,dx$$

$\therefore F_1 = x,\ F_2 = y,\ f_3 = z$

$$\bar{F} = F_1\bar{i} + F_2\bar{j} + F_3\bar{k}$$

$$= x\bar{i} + y\bar{j} + z\bar{k}$$

By Gauss divergence theorem:

$$\iint_S \bar{F}\cdot\bar{n}\,ds = \int_V \operatorname{div}\bar{F}\,dv$$

$$\operatorname{div}\bar{F} = \nabla\bar{F} = \left(\bar{i}\frac{\partial}{\partial x} + \bar{j}\frac{\partial}{\partial y} + \bar{k}\frac{\partial}{\partial z}\right)(x\bar{i} + y\bar{j} + z\bar{k})$$

$$= \frac{\partial}{\partial x}(x) + \frac{\partial}{\partial y}(y) + \frac{\partial}{\partial z}(z)$$

$$= (1+1+1)$$

$$= 3$$

By divergence theorem

$$\iint_S \bar{F}\cdot\bar{n}\,ds = \int_V \operatorname{div}\bar{F}\,dv$$

$$\iint_S F_1\,dy\,dz + f_2\,dx\,dz + f_3\,dx\,dy = \int_V \operatorname{div}\bar{F}\,dv$$

$$\iint_S x\,dy\,dz + y\,dx\,dz + z\,dx\,dy = 3\int_V 1\,dv = 3V$$

$= 3$ (volume of a sphere $x^2+y^2+z^2=a^2$)

$= 3 \cdot \frac{4}{3}\pi(a)^3$

$= 4\pi a^3$

3. State the following vector integral theorems.

(i) Green's theorem:

Let r be the closed region in xy-plane bounded by the curve c if M, N are continuously differentiable functions then

$$\int_C M\,dx + N\,dy = \iint_S \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) dx\,dy$$

(ii) Stoke's theorem:-

'S' be the surface bounded by a closed non intersecting curve 'c'. $\bar{F}$ be any differentiable vector function, then

$$\oint_C \bar{F}\cdot d\bar{r} = \int curl\,\bar{F}\cdot\bar{n}\,ds$$

(iii) Gauss Divergence theorem:-

Let 'S' the surface, enclosed by a volume V, $\bar{F}$ be any differentiable vector function.

then $\int_S \bar{F}\cdot\bar{n}\,ds = \int_V div\,\bar{F}\,d\bar{v}$

⑥ Evaluate $\int_V \bar{F}\,dv$, given that $\bar{F} = 2xz\bar{i} - x\bar{j} + y^2\bar{k}$ and v is the region bounded by the surfaces $x=0, x=2; y=0, y=6$ and $z=x^2, z=4$

Soln

$$= \int_{x=0}^{2}\int_{y=0}^{6}\int_{z=x^2}^{4} (2xz\bar{i} - x\bar{j} + y^2\bar{k})\,dz\,dy\,dx$$

$$= \int_{x=0}^{2}\int_{y=0}^{6} -\left[\frac{2xz^2}{2}\bar{i} - xz\bar{j} + y^2 z\right]_{x^2}^{4} dy\,dx$$

$$= \int_{x=0}^{2}\int_{y=0}^{6} -\{x(x^2)^2\bar{i} - x(x^2)\bar{j} - y^2(x^2) - [x(4)^2\bar{i} - x(4)\bar{j} + y^2(4)]\}$$

$$= \int_{x=0}^{2}\int_{y=0}^{6} -[x^5\bar{i} - x^3\bar{j} + x^2y^2k - 16x\bar{i} + 4x\bar{j} + 4y^2\bar{k}]\,dy\,dx$$

$$= \int_{x=0}^{2}\int_{y=0}^{6} [(16x - x^5)\bar{i} + (x^3 - 4x)\bar{j} + (4y^2 - x^2y^2)\bar{k}]\,dy\,dx$$

$$= \int_{x=0}^{2} (16x - x^5)y\,\bar{i} + (x^3 - 4x)y\,\bar{j} + \left(4\frac{y^3}{3} - x^2\frac{y^3}{3}\right)\bar{k}\Big]_0^6\,dx$$

$$= \int_{x=0}^{2} \{(16x - x^5)6\bar{i} + (x^3 - 4x)6\bar{j} + \left(4\frac{(6)^3}{3} - x^2\frac{(6)^3}{3}\right)\bar{k} - [0+0+0]\}\,dx$$

$$= \int_{x=0}^{2} (96x - 6x^5)\bar{i} + (6x^3 - 24x)\bar{j} + (288 - 72x^2)\bar{k}\,dx$$

$$= \left[\frac{96x^2}{2} - \frac{6x^6}{6}\right]_0^2\bar{i} + \left[\frac{6x^4}{4_2} - \frac{24x^2}{2}\right]_0^2\bar{j} + \left[288x - 72\frac{x^3}{3}\right]_0^2\bar{k}$$

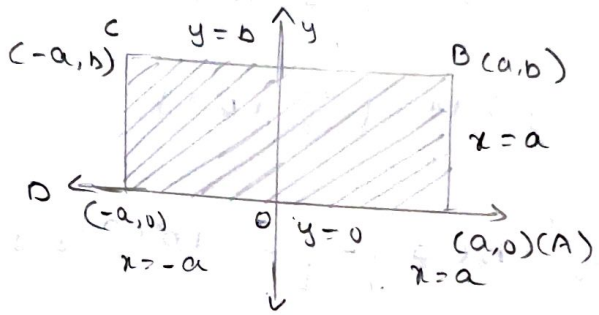
$$= 48(2)^2 - (2)^6]_0^2\,\bar{i} + \left(\frac{3(2)^4}{2} - 12(2)^2 - 0\right)\bar{j} + \left(288(2) - 72\frac{(2)^3}{3}\right)\bar{k}$$

$$\Rightarrow \boxed{128\bar{i} - 24\bar{j} + 384\bar{k}}$$

(1) Verify stoke's theorem for $\bar{F} = (x^2+y^2)\bar{i} - 2xy\bar{j}$ taken around the rectangle bounded by the lines $x = \pm a$, $y = 0$ and $y = b$.

Given, rectangle $x = -a$, $x = +a$, $y = 0$, $y = b$

so given rectangle surface in xy-plane.



given $\bar{F} = (x^2+y^2)\bar{i} - 2xy\bar{j}$

Stokes theorem states that,

$$\oint_C \bar{F}\cdot d\bar{r} = \int_S \text{curl}\,\bar{F}\cdot\bar{n}\,ds$$

To find $\int_C \bar{F}\cdot d\bar{r}$

$$\bar{F}\cdot d\bar{r} = [(x^2+y^2)\bar{i} - 2xy\bar{j}]\cdot[dx\bar{i} + dy\bar{j} + 0\bar{k}]$$

$$= (x^2+y^2)\,dx - 2xy\,dy$$

(i) Along OA: $y = 0$, $dy = 0$, $x = -a$ to $+a$

$$\int_{OA} \bar{F}\cdot d\bar{r} = \int_{x=-a}^{a} x^2\,dx = \frac{x^3}{3}\Big]_{-a}^{a} = \frac{a^3}{3} - \frac{(-a)^3}{3}$$

$$\int_{OA} = \frac{2a^3}{3}$$

(ii) Along AB: $x = a$, $dx = 0$

$y = 0$ to b

$$\bar{F}\,d\bar{r} = (x^2+y^2)\,dx - 2xy\,dy$$

$$= (a^2+y^2)(0) - 2(a)y\,dy = -2ay\,dy$$

$$\int_{AB} \bar{F}\cdot d\bar{r} = \int_{y=0}^{b} -2ay\,dy = -2a\frac{y^2}{2}\Big]_0^b$$

$$\Rightarrow -2a\frac{(b)^2}{2} - 0$$

$$= -ab^2$$

(iii) Along BC: $y = b$, $dy = 0$ and $x = a$ to $-a$

$$\bar{F}\cdot d\bar{r} = \int_{x=a}^{-a} (x^2+b^2)\,dx = \left[\frac{x^3}{3} + b^2x\right]_a^{-a}$$

$$= \left\{\left[\frac{(-a)^3}{3} + b^2(-a)\right] - \left[\frac{a^3}{3} + b^2(a)\right]\right\}$$

$$= \frac{-a^3}{3} - ab^2 - \frac{a^3}{3} - ab^2$$

$$\int_{BC} \bar{F}\cdot d\bar{r} = \frac{-2}{3}a^3 - 2ab^2$$

(iv) Along CD: $x = -a \Rightarrow dx = 0$ & $y = b$ to 0

$$\bar{F}\cdot d\bar{r} = (x^2+y^2)\,dx - 2xy\,dy$$

$$= ((-a)^2+y^2)\,0 - 2(-a)y\,dy = 2ay\,dy$$

$$\int_{CD} \bar{F}\cdot d\bar{r} = \int_{y=b}^{0} 2ay\,dy = 2a\frac{y^2}{2}\Big]_b^0 = a(0-b^2)$$

$$\int_{CD} \bar{F}\cdot d\bar{r} = -ab^2$$

$$\Rightarrow \int_C \bar{F}\cdot d\bar{r} = \int_{OA} \bar{F}\cdot d\bar{r} + \int_{AB} \bar{F}\cdot d\bar{r} + \int_{BC} \bar{F}\cdot d\bar{r} + \int_{CD} \bar{F}\cdot d\bar{r}$$

$$= \frac{+2}{3}a^3 - ab^2 + (-ab^2) - \frac{2}{3}a^3 - 2ab^2$$

$$= -4ab^2 \quad \text{---} \quad (i)$$

To find $\int_C \text{curl}\,\bar{F}\cdot\bar{n}\,ds$

$\bar{F} = (x^2+y^2)\bar{i} - 2xy\,\bar{j}$

$$\text{curl}\,\bar{F} = \nabla\times\bar{F} = \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2+y^2 & -2xy & 0 \end{vmatrix}$$

$$= \bar{i}\left[\frac{\partial}{\partial y}(0) - \frac{\partial}{\partial z}(-2xy)\right] - \bar{j}\left[\frac{\partial}{\partial y}(0) - \frac{\partial}{\partial z}(x^2+y^2)\right] + \bar{k}\left[\frac{\partial}{\partial x}(-2xy) - \frac{\partial}{\partial y}(x^2+y^2)\right]$$

$$= \bar{i}(0-0) - \bar{j}(0-0) + \bar{k}(-2y - (0+2y^2))$$

$$= -4y\bar{k}$$

since curve is in xy plane then normal vector

$\bar{n} = \bar{k}$

$\text{curl}\,\bar{F}\cdot\bar{n} = -4y\,\bar{k}\cdot\bar{k}$

since, $ds = \frac{dx\,dy}{|\bar{n}\cdot\bar{k}|} = \frac{dx\,dy}{|\bar{k}\cdot\bar{k}|} = dx\,dy$

$y=0$, $y=b$ & $x=-a$ to $+a$

Now, $\int_C \text{curl}\,\bar{F}\cdot\bar{n}\,ds = \int_{x=-a}^{a}\int_{y=0}^{b} -4y\,dy\,dx$

$$= -\int_{x=-a}^{a} 4\frac{y^2}{2}\Big]_0^b dx = \int_{x=-a}^{a} -2(b)^2 - 0\,dx$$

$$\Rightarrow -2b^2x\Big]_{-a}^{a} = -2b^2(a) - (-2b^2(-a))$$

$$= -4ab^2 \quad \text{---} (2) \quad \text{By (1) \& (2)}$$

$\int_C \bar{F}\cdot d\bar{r} = \int_S \text{curl}\,\bar{F}\,\bar{n}\,ds$

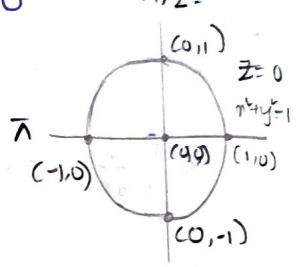
(10) Verify Stoke's theorem for the vector field $\bar{F} = (2x-y)\bar{i} - (yz^2)\bar{j} - y^2z\bar{k}$ over the upper half of the sphere $x^2+y^2+z^2=1$ bounded by its projection on the xy-plane.

Soln Given curve, $x^2+y^2+z^2=1$

the surface is in xy-plane $z=0$

$$x^2+y^2+z^2=1$$

$$\boxed{x^2+y^2=1}$$



$\theta = \frac{3\pi}{2}$

$$\bar{F} = (2x-y)\bar{i} - (yz)\bar{j} - y^2z\bar{k}$$

$$= (2x-y)\bar{i} - (y)(0)\bar{j} - y^2(0)\bar{k}$$

$$\bar{F} = (2x-y)\bar{i}$$

$x = r\cos\theta$, $y = r\sin\theta$

$x = (1)\cos\theta$, $y = (1)\sin\theta$

$dx = -\sin\theta\, d\theta$

Stoke theorem states that

$$\oint_C \bar{F}\cdot d\bar{r} = \int_S \text{curl}\,\bar{F}\cdot\bar{n}\, ds$$

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$$\oint_C \bar{F}\cdot d\bar{r} = \int (2x-y)\, dx$$

$$= \int_{\theta=0}^{2\pi} [2\cos\theta - \sin\theta](-(\sin\theta))\, d\theta$$

$$= \int_{\theta=0}^{2\pi} [-2\sin\theta\cos\theta + \sin^2\theta]\, d\theta$$

$$= \int_{\theta=0}^{2\pi} -\sin 2\theta + \left[\frac{1-\cos 2\theta}{2}\right] d\theta$$

$$= \left\{-\left(\frac{-\cos 2\theta}{2}\right) + \frac{1}{2}\left(\theta - \frac{\sin 2\theta}{2}\right)\right\}_0^{2\pi}$$

$$= \frac{\cos 4\pi}{2} + \frac{1}{2}\left[2\pi - \frac{\sin 4\pi}{2}\right] - \left(\frac{1}{2}\right)$$

$= \frac{1}{2} + \frac{1}{2}\left[(2\pi - 0) - \frac{1}{2}\right]$

$= \pi$

R.H.S $\iint_S \text{curl}\,\bar{F} \cdot \bar{n}\, ds$

$\text{curl}\,\bar{F} = \nabla \times \bar{F}$

$$= \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2x-y & -yz^2 & -y^2z \end{vmatrix}$$

$= \bar{i}\left(\frac{\partial}{\partial y}(-y^2z) - \frac{\partial}{\partial z}(-yz^2)\right) - \bar{j}\left[\frac{\partial}{\partial x}(-y^2z) - \frac{\partial}{\partial z}(2x-y)\right] +$

$\bar{k}\left[\frac{\partial}{\partial x}(-yz^2) - \frac{\partial}{\partial y}(2x-y)\right]$

$= \bar{i}(-2yz + 2yz) - \bar{j}(0+0) - \bar{k}(0-(0-1))$

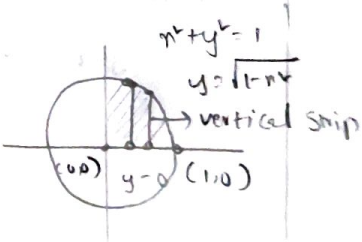
$= \bar{k}$

Since the curve projection is in xy-plane

so, the normal vector $\bar{n} = \bar{k}$

$\therefore \text{curl}\,\bar{F} \cdot \bar{n} = \bar{k} \cdot \bar{k} = 1$

$ds = \frac{dx\,dy}{|\bar{n} \cdot \bar{k}|} = \frac{dx\,dy}{|\bar{k} \cdot \bar{k}|} = dx\,dy$



Circle, $x^2+y^2=1$

Since circle has four equal parts, in the 1st quadrant, $y = 0$ to $\sqrt{1-x^2}$

$x = 0$ to 1

$\therefore \int_C \text{curl}\,\bar{F} \cdot \bar{n}\, ds = 4 \iint \text{curl}\,\bar{F} \cdot \bar{n}\, ds$

(1st quadrant)

$$= 4\int_{x=0}^{1}\int_{y\geq 0}^{\sqrt{1-x^2}} \vec{e}\cdot\vec{e}\, dx\, dy$$

$$= 4\int_{x=0}^{1}\int_{y=0}^{\sqrt{1-x^2}} (1)\, dy\, dx$$

$$= 4\times\int_{x=0}^{1} y\Big]_0^{\sqrt{1-x^2}} dx$$

$$= 4\times\int_{x=0}^{1} [\sqrt{1-x^2}-0]\, dx$$

$$= 4\times\left[\frac{x}{2}\sqrt{1-x^2}+\frac{1}{2}\sin^{-1}\frac{x}{1}\right]_0^1$$

$$= 4\left[0\times\frac{1}{2}\sin^{-1}(1)-0\right]$$

$$= 4\left[\frac{1}{2}\times\frac{\pi}{2}\right]$$

$$= \pi$$

∴ Stoke's theorem is verified

∴ $\int\sqrt{a^2-x^2}\, dx = \frac{x}{2}\sqrt{a^2-x^2}+\frac{a^2}{2}\sin^{-1}\frac{x}{a}$